

GROUP LEADER- MARIAL

Phase 1 – Planning & Design (Week 10) Created the GitHub repository with the correct naming convention and public access. Set up the initial folder structure (/schema, /erd, /documentation, /data). Added all group members and the instructor as collaborators. Drafted the initial README.md with group member roles. Coordinated role assignments and ensured each member understood their Phase 1 deliverables. Merged initial branches and submitted the GitHub link and hardcopy to the LMS. Phase 2 – Schema & Data Population (Week 11) Ensured the /data folder was created for the 100k+ row dataset. Coordinated the dataset generation (Mary) and schema verification (Francheska). Resolved issues when the dataset was regenerated and table names changed (sales → orders, sale_items → order_items). Merged all Phase 2 branches and committed the final message: "100k dataset integrated and verified". Assisted with the Week 11 LMS submission by creating a submission folder and taking a screenshot. Phase 3 – BI Queries & Data Warehousing (Week 12) Created the /queries folder and guided Mary in updating bi_queries.sql to use the new table names. Helped Francheska produce the star schema SQL and diagram. Coordinated with Akisha to re-run row count queries and performance baselines after the dataset change. Assisted Rafael in updating insight_report.pdf and README.md to reflect the corrected queries. Merged all Phase 3 branches after conflicts were resolved. Phase 4 – Cloud Backup & Emerging Tech (Week 13) Assigned tasks for backup/restore (Francheska), cloud deployment (Mary), and reflection essays (Christofer). Resolved merge conflicts on Rafael's branch by instructing him to remove the conflicting PDF temporarily. Created the /backup and /cloud folders and ensured backup.sql, restore_proof.png, and cloud_deployment.pdf were in place. Submitted the Week 13 deliverables to the LMS using a screenshot of the submission folder. Phase 5 – Optimization & Performance Tuning (Week 14) Tasked Mary with providing baseline performance data and Francheska with adding indexes. Asked Rafael to write the optimization_report.pdf and Akisha to verify the results. Created the /optimization folder and collected all required files: original_optimized_queries.sql, explain_before.png, explain_after.png, etc. Merged all optimization-related branches and ensured the final commit reflected Phase 5 completion. General contributions across all phases: Maintained the GitHub repository's health (cleaned up stray folders, resolved merge conflicts, enforced proper branch naming). Acted as the primary point of contact between the group and the instructor. Compiled weekly submission folders and took screenshots for LMS uploads. Kept the team on schedule by setting deadlines and following up on pending tasks.

SQL- MARY

Phase 1 – ERD I designed the Entity Relationship Diagram (ERD) to visually map out the database structure, including tables, relationships, and primary/foreign keys. This served as the blueprint for our inventory and sales system. Phase 2 – Dataset I generated a dataset with over 100,000 rows to populate our database tables with realistic data. This large dataset allowed us to test query performance and simulate real-world database conditions. Phase 3 – Advanced Queries I created five advanced analytical queries for the inventory_sales_db, including monthly revenue trends, top-selling products, and customer spending analysis. These queries were designed to extract meaningful business insights from the database. Phase 4 – Deployment & Backup I assisted Francheska with database backup and restore procedures, then helped deploy our database to Railway.app for cloud hosting. This ensured our database was accessible online and protected against data loss. Phase 5 – Query Optimization I identified 3-5 slow-performing queries from our BI queries, ran EXPLAIN to analyze their execution plans, and recorded baseline performance metrics. I documented the findings in baseline_performance.txt along with EXPLAIN screenshots, then pushed everything to GitHub.

DBA- FAITH

Phase 1 – Database Design & Normalization The ER diagram was co-created with the System Developer, and the initial schema.sql was written containing five tables with primary and foreign keys. Normalization up to 3NF was performed to eliminate redundancy and ensure data integrity. All design outputs were prepared for branch-based version control. Phase 2 – Schema Implementation & Constraint Testing The schema.sql script was executed in XAMPP/phpMyAdmin, successfully creating all tables and verifying key constraints. Referential integrity was tested by inserting a record with a non-existent customer ID, which correctly failed. Phase 3 – Star Schema for BI Reporting A star schema was designed in a separate database, consisting of one fact table and three dimension tables: dim_date, dim_customer, and dim_product. The SQL script and star schema diagram were saved to the /queries/ folder and merged to the main branch. Phase 4 – Backup & Restore A database backup was created using mysqldump, followed by a simulated database drop. Restoration was successfully performed, and COUNT queries verified data integrity, with proof screenshot saved to /backup/. Phase 5 – Indexing & Query Optimization Baseline queries were reviewed, and existing indexes were verified while missing indexes were created as needed. Optimized execution times were recorded in optimized_performance.txt, and EXPLAIN screenshots confirmed improved access types.

TOPE-

Phase 1: Access Control Documentation Role-Based Access (RBAC): Formally documented the system's role-based access protocols to define how different user levels interact with inventory data. Deliverable: Produced the technical specification file: /documentation/security_roles.txt. Phase 2: Database Architecture Reflection NoSQL Analysis: Evaluated the implications of non-relational data structures in the context of system security and flexibility. Deliverable: Authored the nosql_reflection.pdf essay. Phase 3: Data Privacy & Business Intelligence Query Review: Audited Business Intelligence (BI) queries to ensure that data retrieval processes did not violate privacy standards. Insight Reporting: Added critical security notes to the group's insight report to highlight potential privacy risks in data visualization. Phase 4: Emerging Technologies Assessment Security Innovation: Researched and wrote about how modern technological trends impact the security landscape of management systems. Deliverable: Completed the emerging_tech_reflection.pdf document. Phase 5: Optimization & Index Security Index Auditing: Reviewed database indexes created by the team to ensure sensitive columns (such as password_hash or credit_card) were not unnecessarily exposed or indexed. Verification: Confirmed that indexes on product_id, order_date, and order_id were appropriate and did not leak sensitive information. Technical Execution: Documented the final security review in security_review_phase5.txt within the /optimization/ directory. Managed the version control process by committing and pushing these security updates to the tope branch.

DOCUMENTATION LEAD- RAFAEL

Phase 1: Requirements & Document Control Document Quality Assurance: Served as the final reviewer for all Phase 1 deliverables to ensure formatting consistency and professional presentation before submission. Archive Management: Organized the initial project folder structure to ensure all requirements and early-stage drafts were properly tracked. Phase 2: Repository Foundations README Initialization: Authored and formatted the project's README.md file in the GitHub root directory to provide a clear project overview and navigation guide. Documentation Updates: Kept the project documentation synchronized with the evolving system requirements, ensuring the repository remained the "single source of truth" for the team. Phase 3: Design Architecture Documentation Diagram Consolidation: Gathered UML diagrams (Class, Sequence, Deployment) from the design team and integrated them into the formal technical documentation. Layout Standardization: Ensured that all architectural design documents followed a uniform template for professional grading. Phase 4: Optimization Reporting Performance Documentation: Compiled the Optimization Report, gathering the

"Before" and "After" performance data and SQL snapshots from the developers.Folder Governance: Managed the /optimization/ folder on GitHub, ensuring all SQL scripts and performance snapshots were correctly labeled and uploaded.Phase 5: Final Packaging & SubmissionFinal Project Compilation: Assembled the comprehensive Group_FinalProject_Documentation.pdf, merging all phases into the final executive deliverable.Audit & Delivery: Conducted the final check of the public GitHub repository link and commit history to ensure the project met all submission criteria.